

Math 54 Discussion Worksheet

Name: _____ ID: _____

Problem 1

Prove that the complex eigenvalues of real matrices come in conjugate pairs, i.e. $\lambda \in \mathbb{C}$ iff $\bar{\lambda} \in \mathbb{C}$.

Problem 2

1. If $v_1, \dots, v_m$ form a basis for a subspace W of V and x is orthogonal to each v_i , then $x \in W^\perp$.
2. Fix a vector $v \in V$. Define $W_v = \{w : w \cdot v = 0\}$. Show that W_v is a subspace of V .
3. Let W be any subspace of V . Show that $W^\perp$ is a subspace of V .
4. Prove that $\dim(W) + \dim(W^\perp) = \dim(V)$.

Problem 3

Verify the Parallelogram law for vectors $u, v \in \mathbb{R}^n$:

$$\|u + v\|^2 + \|u - v\|^2 = 2\|u\|^2 + 2\|v\|^2.$$